Day 20. Advice for applying machine learning

Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

But it makes unacceptably large errors in predictions. What do you try next?

- Get more training examples
- Try smaller sets of features
- Try getting additional features
- Try adding polynomial features $(x_1^2, x_2^2, x_1x_2, etc)$
- Try decreasing λ
- Try increasing λ

In this week we are going to learn diagnostics of machine learning algorithms,

Machine learning diagnostic

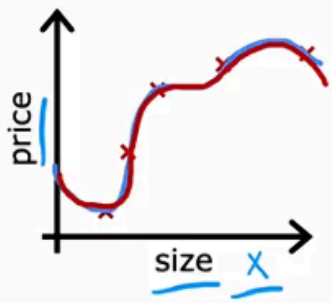
Diagnostic:

A test that you run to gain insight into what is/isn't working with a learning algorithm, to gain guidance into improving its performance.

Diagnostics can take time to implement but doing so can be a very good use of your time.

Evaluating a model

Evaluating your model



Model fits the training data well but will fail to generalize to new examples not in the training set.

x_1 = size in feet^2
x_2 = no. of bedrooms
x_3 = no. of floors
x_4 = age of home in years

$x = x_1$

$W_4 X^4$

$$f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + \cdots + w_n x^n + b$$

just X

we can plot f(x) when we only have one dimention. how do we know the model is overfit if we have more dimentions?

Evaluating your model

Dataset:

	size	price
70%	2104	400
	1600	330
	2400	369
	1416	232
	3000	540
	1985	300
	1534	315
30%	1427	199
	1380	212
	1494	243

training set →

m_{train} = no. training examples

=7

$(x^{(1)}, y^{(1)})$
$(x^{(2)}, y^{(2)})$
$\vdots$
$(x^{(m_{train})}, y^{(m_{train})})$

test set →

$(x_{test}^{(1)}, y_{test}^{(1)})$
$\vdots$
$(x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$

Train/test procedure for linear regression (with squared error cost)

Fit parameters by minimizing cost function $J(\vec{w}, b)$

$$J(\vec{w}, b) = \left[\frac{1}{2m_{train}} \sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m_{train}} \sum_{j=1}^{n} w_j^2\right]$$

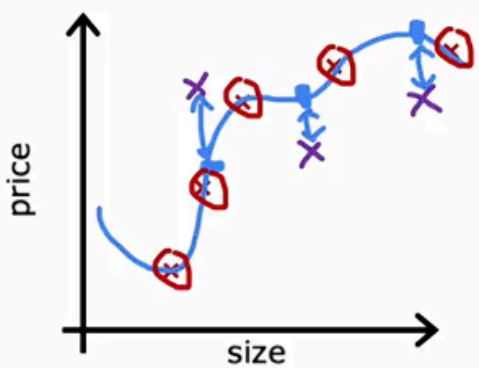
Compute test error:

$$J_{test}(\vec{w}, b) = \frac{1}{2m_{test}} \left[\sum_{i=1}^{m_{test}} \left(f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right) - y_{test}^{(i)}\right)^2\right] \quad \cancel{\sum_{j=1}^{n} w_j^2}$$

Compute training error:

$$J_{train}(\vec{w}, b) = \frac{1}{2m_{train}} \left[\sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}\left(\vec{x}_{train}^{(i)}\right) - y_{train}^{(i)}\right)^2\right]$$

Train/test procedure for linear regression (with squared error cost)



X = train

X = test

$J_{train}(\vec{w}, b)$ will be low

$J_{test}(\vec{w}, b)$ will be high

then we can find whether the results have generalization or not.

Train/test procedure for classification problem

0/1

Fit parameters by minimizing $J(\vec{w}, b)$ to find $\vec{w}, b$

E.g.,

$$J(\vec{w}, b) = -\frac{1}{m_{train}} \sum_{i=1}^{m_{train}} \left[y^{(i)} \log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1 - y^{(i)}) \log\left(1 - f_{\vec{w},b}(\vec{x}^{(i)})\right) \right] + \frac{\lambda}{2m_{train}} \sum_{j=1}^{n} w_j^2$$

Compute test error:

$$J_{test}(\vec{w}, b) = -\frac{1}{m_{test}} \sum_{i=1}^{m_{test}} \left[y_{test}^{(i)} \log\left(f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right)\right) + \left(1 - y_{test}^{(i)}\right) \log\left(1 - f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right)\right) \right]$$

Compute train error:

$$J_{train}(\vec{w}, b) = -\frac{1}{m_{train}} \sum_{i=1}^{m_{train}} \left[y_{train}^{(i)} \log\left(f_{\vec{w},b}\left(\vec{x}_{train}^{(i)}\right)\right) + \left(1 - y_{train}^{(i)}\right) \log\left(1 - f_{\vec{w},b}\left(\vec{x}_{train}^{(i)}\right)\right) \right]$$

Train/test procedure for classification problem

fraction of the test set and the fraction of the train set that the algorithm has misclassified.

$$\hat{y} = \begin{cases} 1 \text{ if } f_{\vec{w},b}(\vec{x}^{(i)}) \geq 0.5 \\ 0 \text{ if } f_{\vec{w},b}(\vec{x}^{(i)}) < 0.5 \end{cases}$$

count $\hat{y} \neq y$

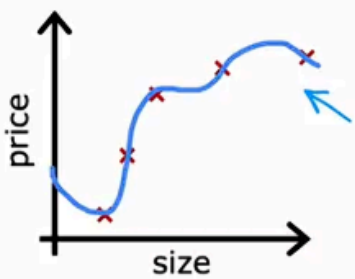
$J_{test}(\vec{w}, b)$ is the fraction of the test set that has been misclassified.

$J_{train}(\vec{w}, b)$ is the fraction of the train set that has been misclassified.

with this, we are a step toward a automatic algorithm that can help us choose the polynomiom we are going to use in the function.

Model selection and training/cross validation/test sets

Model selection (choosing a model)



Once parameters $\vec{w}, b$ are fit to the training set, the training error $J_{train}(\vec{w}, b)$ is likely lower than the actual generalization error.

$J_{test}(\vec{w}, b)$ is better estimate of how well the model will generalize to new data compared to $J_{train}(\vec{w}, b)$.

$x = x_1$

$f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$

Model selection (choosing a model)

$d=1$ 1. $f_{\vec{w},b}(\vec{x}) = w_1 x + b$ $\rightarrow w^{<1>}, b^{<1>} \rightarrow J_{test}(w^{<1>}, b^{<1>})$

$d=2$ 2. $f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + b$ $\rightarrow w^{<2>}, b^{<2>} \rightarrow J_{test(w^{<2>},b^{<2>})}$

$d=3$ 3. $f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + w_3 x^3 + b$ $\rightarrow w^{<3>}, b^{<3>} \rightarrow J_{test(w^{<3>},b^{<3>})}$

$\vdots$

$d=10$ 10. $f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + \cdots + w_{10} x^{10} + b$ $\rightarrow J_{test(w^{<10>},b^{<10>})}$

Choose $w_1 x + \cdots + w_5 x^5 + b$ $d=5$ $J_{test}(w^{<5>}, b^{<5>})$

How well does the model perform? Report test set error $J_{test}(w^{<5>}, b^{<5>})$?

The problem: $J_{test}(w^{<5>}, b^{<5>})$ is likely to be an optimistic estimate of generalization error (ie. $J_{test}(w^{<5>}, b^{<5>}) <$ generalization error). Because an extra parameter d (degree of polynomial) was chosen using the test set.

> This is because when you use the test set to select a model (such as choosing the order of a polynomial d), you are effectively "tuning parameters using the test set". The test set is originally intended to evaluate the performance of the final model, but if it is used to determine which model is optimal, the test set error will no longer serve as an independent evaluation metric. It is like an exam: if you look at the answers in advance, your exam score will naturally be higher than your actual ability. As a result, the test error will be underestimated and cannot truly reflect the model's performance on unseen data.

Training/cross validation/test set

size	price
2104	400
1600	330
2400	369
1416	232
3000	540
1985	300
1534	315
1427	199
1380	212
1494	243

training set 60% → $(x^{(1)}, y^{(1)}) \ \vdots \ (x^{(m_{train})}, y^{(m_{train})})$ $m_{train} = 6$

cross validation 20% → $(x_{cv}^{(1)}, y_{cv}^{(1)}) \ \vdots \ (x_{cv}^{(m_{cv})}, y_{cv}^{(m_{cv})})$ $m_{cv} = 2$

test set 20% → $(x_{test}^{(1)}, y_{test}^{(1)}) \ \vdots \ (x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$ $m_{test} = 2$

we also call cv set dev set, but cross validation is a more common name.

Training/cross validation/test set

Training error: $$J_{train}(\vec{w}, b) = \frac{1}{2m_{train}}\left[\sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2\right]$$

Cross validation error: $$J_{cv}(\vec{w}, b) = \frac{1}{2m_{cv}}\left[\sum_{i=1}^{m_{cv}} \left(f_{\vec{w},b}\left(\vec{x}_{cv}^{(i)}\right) - y_{cv}^{(i)}\right)^2\right]$$ (validation error, dev error)

Test error: $$J_{test}(\vec{w}, b) = \frac{1}{2m_{test}}\left[\sum_{i=1}^{m_{test}} \left(f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right) - y_{test}^{(i)}\right)^2\right]$$

of course, these formulas do not include regularzation term since it will be used only in training process.

Model selection

$d=1$ 1. $f_{\vec{w},b}(\vec{x}) = w_1 x + b$ $\quad w^{\langle 1\rangle}, b^{\langle 1\rangle} \rightarrow J_{cv}(w^{\langle 1\rangle}, b^{\langle 1\rangle})$

$d=2$ 2. $f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + b$ $\quad \rightarrow J_{cv}(w^{\langle 2\rangle}, b^{\langle 2\rangle})$

$d=3$ 3. $f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + w_3 x^3 + b$

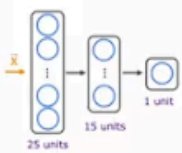
$\vdots$ $\quad \vdots \quad \vdots$

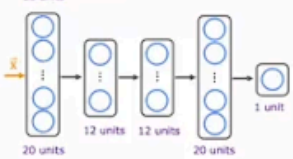
$d=10$ 10. $f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + \cdots + w_{10} x^{10} + b$ $\quad J_{cv}(w^{\langle 10\rangle}, b^{\langle 10\rangle})$

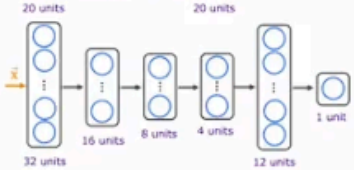
$\rightarrow$ Pick $w_1 x + \cdots + w_4 x^4 + b$ $\quad (J_{cv}(w^{<4>}, b^{<4>}))$

Estimate generalization error using test the set: $J_{test}(w^{<4>}, b^{<4>})$

Model selection – choosing a neural network architecture

1. 

2. 

3. 

Pick $\boldsymbol{w}^{<2>}, \boldsymbol{b}^{<2>}$

Estimate generalization error using the test set: $J_{test}(\boldsymbol{w}^{<2>}, \boldsymbol{b}^{(<2>)})$